

Mathematics — Diagnostic 1 · Answer key and marking

For the parent. Mark each question ✓ or ✗, then fill the skill table on the last page and upload a photo of it (or of the whole test).

A. Numbers you learned last year

1. [G4.1 · tier 1] 406,029
2. [G4.1 · tier 2] 740,000
3. [G4.2 · tier 2] 32,448
4. [G4.3 · tier 1] 56, 54, 84, 32, 81, 132 (time this row: should be under 20 s)
5. [G4.4 · tier 2] 22,722
6. [G4.4 · tier 2] 1,058
7. [G4.5 · tier 2] 492 r 5
8. [G4.6 · tier 2] 1,2,3,4,6,9,12,18,36 — primes: 2, 3
9. [G4.7 · tier 2] $\frac{3}{4}$ ($= \frac{6}{8} > \frac{5}{8}$)
10. [G4.7 · tier 1] $\frac{3}{6} = \frac{1}{2}$
11. [G4.8 · tier 2] $\frac{47}{100}$; 0.5 is bigger
12. [G4.9 · tier 3] $24 \times 15 = 360$; $9 \times 32 = 288$; $360 - 288 = 72$

B. Place value and powers of ten

13. [E02.1 · tier 1] 500; 5,000; 10
14. [E02.2 · tier 2] Two billion, forty-six million, three hundred thousand, fifteen
15. [E02.2 · tier 2] $800,000,000 + 30,000,000 + 200,000 + 7,000$
16. [E02.2 · tier 2] $4,009,000 < 4,090,000 < 4,099,000 < 4,900,000$
17. [E02.4 · tier 1] $10^4 = 10,000$
18. [E02.4 · tier 2] 3,700; 6,000; 45
19. [E02.4 · tier 2] 12,000 ($3 \times 4 = 12$, then three zeros)

C. Multiplication, division and properties

20. [E02.3 · tier 2] 5; associative property
21. [E02.3 · tier 2] $6 \times 40 + 6 \times 8 = 240 + 48 = 288$
22. [E02.5 · tier 3] 173,700
23. [E02.6 · tier 2] $651 \times 8 = 5,208$ — yes

D. Expressions and order of operations

24. [E02.7 · tier 2] $(12 + 8) \times 3$
25. [E02.7 · tier 2] Six times five, subtracted from forty (or similar)
26. [E02.8 · tier 1] 18
27. [E02.8 · tier 2] 30
28. [E02.8 · tier 3] 17

29. [E01.2 · tier 2] He added first; multiply first: $5 + 6 = 11$

Skill scorecard

Skill	Description	Questions	Correct / total	Status
G4.1	Read, write, compare and round whole numbers to 1,000,000	1, 2	___ / 2	
G4.2	Add and subtract multi-digit numbers with regrouping	3	___ / 1	
G4.3	Multiplication facts to 12 x 12 (fluent, under 3 s each)	4	___ / 1	
G4.4	Multiply 3–4 digit by 1 digit and 2 digit by 2 digit	5, 6	___ / 2	
G4.5	Divide up to 4 digit by 1 digit, including remainders	7	___ / 1	
G4.6	Factors, multiples, prime and composite numbers	8	___ / 1	
G4.7	Equivalent fractions; compare; add/subtract like denominators	9, 10	___ / 2	
G4.8	Decimals to hundredths; relate to fractions; compare	11	___ / 1	
G4.9	Multi-step word problems with all four operations	12	___ / 1	
E01.2	Explain reasoning in words using correct maths vocabulary	29	___ / 1	
E02.1	Place-value patterns: a digit is 10x the place to its right, 1/10 of the place to its left	13	___ / 1	
E02.2	Read, write and compare whole numbers to billions (standard, word, expanded form)	14, 15, 16	___ / 3	
E02.3	Properties of operations: commutative, associative, distributive	20, 21	___ / 2	
E02.4	Powers of 10 and exponents; multiply by 10, 100, 1000 using patterns	17, 18, 19	___ / 3	
E02.5	Multiply by 1-digit and 2-digit numbers (standard algorithm)	22	___ / 1	
E02.6	Relate multiplication and division; use inverse to check	23	___ / 1	
E02.7	Write a numerical expression from words and vice versa	24, 25	___ / 2	
E02.8	Evaluate expressions using order of operations, including parentheses and brackets	26, 27, 28	___ / 3	

Status: all correct = **Secure** · half or more = **Emerging** · less than half = **Not yet**. Tier 1 = basic, 2 = expected mastery, 3 = A-level. Also note total minutes taken and whether Q4 (times tables) took more than 20 seconds.